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66
67
```

Part 1: Variables, Data Types, and Basic Math

Python Variables

```
In [1]: 1 #Question 1: Create a variable named age, assign it the value 25, and print it
        2 age=25
        3 print(age)
```

25

```
In [14]: 1 #Question 2: Swap the values of two variables x = 5 and y = 10 without using a temporary variable
        2 x=5
        3 y=10
        4 x,y=y,x
        5 print("Value of x is:",x)
        6 print("Value of y is:",y)
```

Value of x is: 10

Value of y is: 5

```
In [15]: 1 #Question 3: Check the data type of the variable price = 49.99 using a built-in function
        2 price =49.99
        3 type(price)
```

Out[15]: float

Numeric Types & Arithmetic

```
In [29]: 1 #Question 4: Calculate the remainder when 145 is divided by 12.
2 a=145
3 b=12
4 reminder=145%12
5 print("The remainder when 145 is divided by 1:",reminder)
6
7 #Question 5: Raise 5 to the power of 3 using a Python operator.
8 a=5**3
9 print("Raise 5 to the power of 3 using a Python operator:",a)
10
11 #Question 6: Divide 10 by 3 and ensure the result is a whole number (integer)
12 b=10/3
13 c=int(b)
14 print("Divide 10 by 3 and ensure the result is a whole numbe:",c)
15
16 #Question 7: Convert the float value 9.99 into an integer.
17 d=9.99
18 e=int(d)
19 print("Convert the float value 9.99 into an integer:",e)
```

The remainder when 145 is divided by 1: 1

Raise 5 to the power of 3 using a Python operator: 125

Divide 10 by 3 and ensure the result is a whole numbe: 3

Convert the float value 9.99 into an integer: 9

String Operations

```
In [48]: 1 #Question 8: Combine the strings "Data" and "Science" into a single string
2 st1="Data"
3 st2="Science"
4 st3=st1+" "+st2
5 print("Combine ans of two strings is:",st3)
6
7 #Question 9: Extract the first three characters from the string "Python".
8 st1="Python"
9 first_three=st1[:3]
10 print("Extracted first three char are:",first_three)
11
12 #Question 10: Convert the string "machine Learning" to uppercase.
13 st="machine Learning"
14 st_up=st.upper()
15 print("Convert the string machine learning to uppercase is:",st_up)
16
17 #Question 11: Count how many times the letter "a" appears in the word "anaconda"
18 st="anaconda"
19 print("Count how many times the letter a:",st.count("a"))
20
21 #Question 12: Check if the word "Data" exists in the text "Big Data Analytics"
22 st="Big Data Analytics"
23 st1=st.split()
24 print(st1)
25 i="Data"
26 for i in st1:
27     if i=="Data":
28         print("the word Data exists in the text Big Data Analytics")
```

Combine ans of two strings is: Data Science

Extracted first three char are: Pyt

Convert the string machine learning to uppercase is: MACHINE LEARNING

Count how many times the letter a: 3

['Big', 'Data', 'Analytics']

the word Data exists in the text Big Data Analytics

Part 2: Data Structures

Python Lists

```

In [63]: 1 #Question 13: Create a list containing five of your favorite fruits.
2 lst=["Apple","Mango","Orange","Grapes","Strawberry"]
3 print(lst)
4
5 #Question 14: Add the element "orange" to the end of an existing list.
6 lst.append("Orange")
7 print(lst)
8
9 #Question 15: Insert "apple" at the very beginning (index 0) of a list.
10 lst.insert(0,"Apple")
11 print(lst)
12
13 #Question 16: Remove the third item from a list of numbers.
14 lst.pop(3)
15 print(lst)
16
17 #Question 17: Find the length of a list named data_points.
18 data_points=lst
19 print("length of a list named data_points is:",len(data_points))
20
21 #Question 18: Slice a list nums = [10, 20, 30, 40, 50] to get only [20, 30, 40]
22 nums=[10,20,30,40,50]
23 print(nums[1:4])

```

```

['Apple', 'Mango', 'Orange', 'Grapes', 'Strawberry']
['Apple', 'Mango', 'Orange', 'Grapes', 'Strawberry', 'Orange']
['Apple', 'Apple', 'Mango', 'Orange', 'Grapes', 'Strawberry', 'Orange']
['Apple', 'Apple', 'Mango', 'Grapes', 'Strawberry', 'Orange']
length of a list named data_points is: 6
[20, 30, 40]

```

Dictionaries

```

In [70]: 1 #Question 19: Create a dictionary representing a student with keys: "name"
2 student={"name":["Sachin"],
3          "grade":[70],
4          "age":[20]}
5 print(student)

```

```
{'name': ['Sachin'], 'grade': [70], 'age': [20]}
```

```

In [71]: 1 #Question 20: Access the value of the key "grade" from your student dictionary
2 student.get("grade")

```

Out[71]: [70]

```

In [72]: 1 #Question 21: Add a new key-value pair "passing": True to an existing dictionary
2 student["passing"]=True
3 student

```

Out[72]: {'name': ['Sachin'], 'grade': [70], 'age': [20], 'passing': True}

```
In [79]: 1 #Question 22: Delete the key "age" from a dictionary.
        2 student.pop("age")
```

```
In [81]: 1 student
```

```
Out[81]: {'name': ['Sachin'], 'grade': [70], 'passing': True}
```

```
In [83]: 1 #Question 23: Print all the keys present in a dictionary.
        2 student.keys()
```

```
Out[83]: dict_keys(['name', 'grade', 'passing'])
```

Tuples & Sets

```
In [86]: 1 #Question 24: Create a tuple containing three coordinates: (x, y, z).
        2 tup=("x","y","z")
        3 print("Create a tuple containing three coordinates: (x, y, z):",tup)
```

```
Create a tuple containing three coordinates: (x, y, z): ('x', 'y', 'z')
```

```
In [88]: 1 #Question 25: Try to change the first element of a tuple and observe the error.
        2 tup[0]="a"
        3 tup
```

```
-----
TypeError                                Traceback (most recent call last)
~\AppData\Local\Temp\ipykernel_14228\1398568576.py in <module>
      1 #Question 25: Try to change the first element of a tuple and observe
the error. Explain why it happens.
----> 2 tup[0]="a"
      3 tup
```

```
TypeError: 'tuple' object does not support item assignment
```

```
In [ ]: 1 "TypeError: 'tuple' object does not support item assignment"
        2 "The error occurred due to tuple is immutable we can not change tuple once"
```

```
In [89]: 1 #Question 26: Create a list with duplicate numbers and convert it into a set.
        2 lst=[1,2,1,3,4,3,2,4,5,2,4]
        3 print(lst)
        4 st=set(lst)
        5 print(st)
```

```
[1, 2, 1, 3, 4, 3, 2, 4, 5, 2, 4]
{1, 2, 3, 4, 5}
```

```
In [93]: 1 #Question 27: Find the common elements between two sets: {1, 2, 3} and {3, 4, 5}
2 set1={1,2,3}
3 set2={3,4,5}
4 print(set1&set2)
5 print(set1.intersection(set2))

{3}
{3}
```

Part 3: Control Flow and Loops

Conditional Statements

```
In [97]: 1 #Question 28: Write an if statement that prints "Pass" if a score is 50 or above
2 score=int(input("Enter Score:"))
3 if score>=50:
4     print("Pass")
5 else:
6     print("Fail")
7 #Question 29: Write an if-else block to check if an input number is even or odd
8 num=int(input("Enter Number:"))
9 if num%2==0:
10     print("Even")
11 else:
12     print("Odd")
```

Enter Score:50

Pass

Enter Number:4

Even

```
In [101]: 1 #Question 30: Create an if-elif-else chain to classify a temperature as "Cold", "Warm", or "Hot"
2 temp=int(input("Enter Temperature:"))
3 if temp<=15:
4     print("The temperature is 'Cold'")
5 elif 15<temp<=30:
6     print("The temperature is 'Warm'")
7 elif temp>30:
8     print("The temperature is 'Hot'")
```

Enter Temperature:15

The temperature is 'Cold'

Loops

In [114]:

```
1 #Question 31: Write a for loop to print numbers from 1 to 10.
2 for i in range(1,11):
3     print(i)
```

1
2
3
4
5
6
7
8
9
10

In [115]:

```
1 #Question 32: Use a for loop to calculate the sum of all numbers in a list
2 lst=[1,2,3,4,5,6,7,8]
3 sum1=0
4 for i in lst:
5     sum1+=i
6 print("sum of all numbers in a list is:",sum1)
```

sum of all numbers in a list is: 36

In [112]:

```
1 #Question 33: Write a while loop that prints numbers from 5 down to 1.
2 i=5
3 while i>0:
4     print(i)
5     i=i-1
```

5
4
3
2
1

In [123]:

```
1 #Question 34: Create a for loop that iterates through a list but skips the
2 lst=[1,2,3,4,5,6,7,8,9,10]
3 for i in lst:
4     if i==5:
5         continue
6     else:
7         print(i)
8
```

1
2
3
4
6
7
8
9
10

```
In [126]: 1 #Question 35: Exit a while loop immediately when a counter variable hits 1
2 for i in lst:
3     if i==3:
4         break
5     print(i)

1
2
```

Part 4: Functions and Basic Logic

Defining Functions

```
In [128]: 1 #Question 36: Define a function named greet that takes a name as an argument
2 def greet(name):
3     print(f"Welcome to {name} Program")
4
5 greet("Data Science")
```

Welcome to Data Science Program

```
In [130]: 1 #Question 37: Write a function that accepts two numbers and returns their product
2 def product(a,b):
3     product=a*b
4     print("Product of two number is:", product)
5
6 product(5,4)
```

Product of two number is: 20

```
In [142]: 1 #Question 38: Create a function that takes a string and returns its total length
2 def char_len(st):
3     print(f"The length of string is:",len(st))
4
5 str1=input("Enter String:")
6 char_len(str1)
```

Enter String:Data science
The length of string is: 12

```
In [146]: 1 #Question 39: Write a function that returns True if a number is positive, False otherwise
2 def get_num(num):
3     if num>0:
4         print("True")
5     elif num<0:
6         print("False")
7
8 get_num(int(input("Enter Number")))
```

Enter Number23
True

```
In [147]: 1 #Question 40: Define a function with a default parameter value for a city
2 def city_name(city):
3     print(f"The City name is:{city}")
4
5 city_name(input("Enter city name:"))
```

Enter city name:Pune
The City name is:Pune

Part 5: Data Science Foundations (Built-ins & Logic)

Data Math Foundations

```
In [150]: 1 #Question 41: Find the maximum and minimum values in a list of numbers wi
2 lst=[1,43,3,23,46,5,2,12,65,4,32,3]
3 max1=lst[0]
4 min1=lst[0]
5 for i in lst:
6     if i<min1:
7         min1=i
8
9 for i in lst:
10     if i>max1:
11         max1=i
12
13 print(f"minimum no is:{min1}")
14 print(f"maximum no is:{max1}")
```

minimum no is:1
maximum no is:65

```
In [158]: 1 #Question 42: Round the number 3.14159 to exactly two decimal places.
2 num=3.14159
3 print(round(num,2))
4
```

3.14

```
In [157]: 1 #Question 43: Generate a List of integers from 0 to 100 with a step of 10
2 lst=[]
3 for i in range(0,100,10):
4     lst.append(i)
5 print(lst)
```

[0, 10, 20, 30, 40, 50, 60, 70, 80, 90]